

HEMOLYTIC ANEMIAS-LAB INVESTIGATIONS

1. Introduction

- Hemolytic anemia refers to a group of disorders characterized by **premature destruction of red blood cells (RBCs)** before the end of their normal lifespan of about **120 days**.
- When RBC destruction exceeds the capacity of the bone marrow to replace them, anemia develops.

- Hemolysis may occur in two main locations:
- **Extravascular hemolysis**
 - Occurs mainly in the **spleen, liver, and bone marrow**
 - RBCs are destroyed by macrophages
 - Common in immune hemolysis and hereditary spherocytosis
- **Intravascular hemolysis**
 - RBCs are destroyed **within blood vessels**
 - Leads to release of hemoglobin into plasma
 - Seen in conditions such as **malaria, transfusion reactions, paroxysmal nocturnal hemoglobinuria, and mechanical hemolysis**

Feature	Intravascular Hemolysis	Extravascular Hemolysis
Site of RBC destruction	Occurs within blood vessels	Occurs in the reticuloendothelial system , mainly spleen, liver, and bone marrow
Mechanism	RBCs rupture directly in circulation releasing hemoglobin into plasma	RBCs are phagocytosed by macrophages in spleen or liver
Free plasma hemoglobin	Markedly increased	Usually absent or minimal
Hemoglobinemia	Present	Absent
Hemoglobinuria	Present (dark or red urine)	Absent
Hemosiderinuria	Common in chronic cases	Rare
Serum haptoglobin	Markedly decreased or absent due to binding of free hemoglobin	Mildly decreased or normal
Serum LDH	Very high	Moderately elevated
Unconjugated bilirubin	Increased	Increased (often more prominent)
Peripheral blood film findings	Schistocytes (RBC fragments)	Spherocytes
Splenomegaly	Usually absent	Common due to increased RBC destruction
Jaundice	May occur but less prominent	Common due to increased bilirubin production
Common causes	Malaria, transfusion reactions, mechanical heart valves, paroxysmal nocturnal hemoglobinuria, DIC	Hereditary spherocytosis, autoimmune hemolytic anemia, hemoglobinopathies

2. Laboratory Evidence of Hemolysis

- Initial laboratory investigations aim to **confirm that hemolysis is occurring.**
- **2.1 Complete Blood Count (CBC)**
- The **CBC** is usually the first investigation performed.
- Typical findings include:
 - **Reduced hemoglobin (Hb)**
 - **Reduced hematocrit (PCV)**
 - **Increased reticulocyte count**
 - **Increased Mean Corpuscular Volume (MCV)** due to reticulocytosis
- Interpretation:
 - Reticulocytosis indicates **bone marrow compensation for RBC destruction.**

2.2 Reticulocyte Count

- Reticulocytes are immature RBCs released from the bone marrow.
- Findings:
 - **Increased reticulocyte count (>2–3%)**
- Significance:
 - Indicates **increased erythropoietic activity**
 - Helps differentiate **hemolytic anemia from hypoproliferative anemia**
 - However, in severe cases where bone marrow is suppressed, the reticulocyte count may not be elevated.

2.3 Peripheral Blood Film (PBF)

- Microscopic examination of a stained blood smear provides valuable diagnostic clues.
- Characteristic morphological changes include:
- **Spherocytes**
- Small dense RBCs without central pallor
- Seen in:
 - Hereditary spherocytosis
 - Autoimmune hemolytic anemia

- **Schistocytes (fragmented RBCs)**
- Irregular RBC fragments
- Seen in:
 - Microangiopathic hemolytic anemia
 - Disseminated intravascular coagulation
 - Mechanical heart valves

Bite Cells

- RBCs with semicircular defects
- Seen in **G6PD deficiency**

- **Target Cells**

- Seen in:

- Hemoglobinopathies
- Liver disease
- Thalassemia

- **Sickled Cells**
- Seen in **sickle cell disease**
- Peripheral blood film examination is therefore essential in narrowing the differential diagnosis.

3. Biochemical Markers of Hemolysis

- **3.1 Serum Bilirubin**
- During RBC destruction, hemoglobin is broken down into bilirubin.
- Types measured:
- **Unconjugated (indirect) bilirubin**
- **Conjugated (direct) bilirubin**
- Findings in hemolytic anemia:
- **Elevated unconjugated bilirubin**
- Clinical effect:
- May cause **jaundice**.

3.2 Lactate Dehydrogenase (LDH)

- LDH is an intracellular enzyme released during cell destruction.
- Findings:
 - **Elevated LDH levels**
- Significance:
 - Particularly high in **intravascular hemolysis**

3.3 Haptoglobin

- Haptoglobin is a plasma protein that binds free hemoglobin.
- Findings:
 - **Decreased haptoglobin levels**
- Explanation:
 - Free hemoglobin released during hemolysis binds to haptoglobin.
 - The complex is rapidly cleared from circulation.
 - Low haptoglobin is therefore an important indicator of **intravascular hemolysis**.

4. Urine Examination

- Urine analysis may reveal products of hemoglobin breakdown.
- **4.1 Hemoglobinuria**
- Occurs when **free hemoglobin is filtered through the kidneys.**
- Seen in:
 - Severe intravascular hemolysis
 - Acute transfusion reactions
 - Malaria
- Urine appears **reddish or dark brown.**

- **4.2 Hemosiderinuria**

- Occurs when iron from hemoglobin accumulates in renal tubular cells.
- Detected using:
 - **Prussian blue stain**
- Indicates **chronic intravascular hemolysis**.

5. Immunological Tests

- **5.1 Direct Antiglobulin Test (DAT) / Direct Coombs Test**
- This test detects **antibodies or complement attached to RBCs**.
- Principle:
- Anti-human globulin reagent agglutinates RBCs coated with IgG or complement.
- Uses:
- Diagnosis of:
- **Autoimmune hemolytic anemia**
- **Hemolytic disease of the newborn**
- **Hemolytic transfusion reactions**

- interpretation:
- **Positive DAT → immune-mediated hemolysis**

- **5.2 Indirect Antiglobulin Test (IAT)**
- Detects **free antibodies in patient serum.**
- Uses:
 - Antibody screening
 - Crossmatching in blood transfusion.

Specialized Tests

- **6.1 Osmotic Fragility Test**
- Used to diagnose **hereditary spherocytosis**.
- Principle:
- RBCs are placed in hypotonic saline solutions.
- Result:
- Spherocytes lyse **earlier than normal RBCs**.

- **Principle**

- Red blood cells are placed in **hypotonic saline solutions**. Water enters the cells through osmosis, causing them to swell and eventually rupture.
- RBCs with **membrane defects** (such as spherocytes) have **reduced surface area** and therefore rupture more easily.

- **Procedure**

- Prepare a series of **saline solutions with decreasing concentrations** (e.g., 0.85% to 0.10% NaCl).
- Add a small amount of blood to each tube.
- Incubate at room temperature.
- Centrifuge and observe hemolysis.

Condition

Hereditary spherocytosis

Autoimmune hemolytic anemia

Thalassemia

Iron deficiency anemia

Result

Increased osmotic fragility

Increased fragility

Decreased osmotic fragility

Decreased fragility

6.2 G6PD Enzyme Assay

- Used to diagnose **Glucose-6-Phosphate Dehydrogenase deficiency**.
- Principle:
- Measures the activity of the G6PD enzyme in RBCs.
- Significance:
- Deficiency leads to:
- Increased oxidative damage
- Episodic hemolysis
- Triggers include:
- Certain drugs
- Infections
- Fava beans.

Sickling Test

- **Sickling Test Using Sodium Metabisulphite (Sodium Metabisulfite Test)**
- The **sodium metabisulphite sickling test** is a screening laboratory test used to detect the presence of Hemoglobin S (HbS) in blood. It is commonly used in the diagnosis of **sickle cell disease and sickle cell trait**.

- **1. Principle of the Test**

- The test is based on the ability of **sodium metabisulphite to reduce oxygen tension in the blood sample.**
- Sodium metabisulphite acts as a **reducing agent**, removing oxygen from hemoglobin.
- When **Hemoglobin S (HbS)** becomes **deoxygenated**, it **polymerizes**.
- This polymerization causes **red blood cells (RBCs) to change shape from normal biconcave discs to sickle or crescent shapes.**
- These sickled cells can be **observed under a microscope.**
- Normal hemoglobin (**HbA**) does **not sickle** under these conditions.

- **5. Procedure**

- Place **one drop of blood** on a clean microscope slide.
- Add **one drop of 2% sodium metabisulphite solution**.
- Mix gently using a stick or applicator.
- Cover with a **cover slip**.
- **Seal the edges** of the cover slip with petroleum jelly or nail polish to prevent oxygen entry.
- Allow the preparation to **stand for 10–30 minutes**.
- Examine under the **microscope (×40 objective)**.

- **8. Confirmatory Tests**

- Positive screening tests should be confirmed by:
- **Hemoglobin electrophoresis**
- **High performance liquid chromatography (HPLC)**
- **Isoelectric focusing**
- **Molecular testing**

Hemoglobin Electrophoresis

- Hemoglobin electrophoresis is a **laboratory technique used to separate and identify different types of hemoglobin** based on their electrical charge and migration in an electric field.
- It is an essential diagnostic test in hematology for identifying **hemoglobinopathies** and **thalassemias**, which are genetic disorders affecting hemoglobin structure or production.
- Hemoglobin electrophoresis works on the principle that **different hemoglobin molecules carry different electrical charges**, allowing them to migrate at different speeds in a supporting medium when an electric current is applied.

- **2. Structure of Hemoglobin**
- Hemoglobin is a **tetrameric protein** composed of:
- **4 globin chains**
- **4 heme groups**
- Each heme contains **iron (Fe^{2+})** that binds oxygen.

Hemoglobin

Globin Chain Composition

Percentage in Adults

HbA

$\alpha_2\beta_2$

95–98%

HbA₂

$\alpha_2\delta_2$

2–3%

HbF

$\alpha_2\gamma_2$

<1%

- **Abnormal Hemoglobins**

- Common abnormal hemoglobins include:

- HbS – sickle hemoglobin

- HbC

- HbD

- HbE

- These arise from **mutations in globin genes**, altering amino acid sequences and electrical charge.

- **3. Principle of Hemoglobin Electrophoresis**

- Electrophoresis separates molecules according to:

- **Electrical charge**

- **Molecular size**

- **Shape**

- **pH of the buffer system**

- Hemoglobin molecules migrate toward the **anode (+)** or **cathode (-)** depending on their charge.

- At **alkaline pH (8.4–8.6)**:
- Hemoglobin molecules carry a **negative charge** and migrate toward the **anode**.
- Different hemoglobins move at different rates due to differences in **amino acid composition**.

Typed of hemoglobin Elect

- **1. Alkaline Electrophoresis**
- Most commonly used screening method.
- Medium:
- Cellulose acetate membrane
- Buffer:
- Tris-EDTA-borate buffer
- pH 8.4–8.6
- Separates:
- HbA, HbF, HbS, HbC
- Migration pattern:

Hemoglobin

HbA

HbF

HbS

HbC

Relative Migration

Fast

Slightly slower

Slower

Slowest

- **6. Preparation of Hemolysate**

- Before electrophoresis, hemoglobin must be extracted from RBCs.

- **Procedure**

1. Collect EDTA blood

2. Wash red cells with **normal saline**

3. Lyse RBCs using:

1. Distilled water

2. Hemolysing reagent

4. Centrifuge

5. Collect **supernatant hemolysate**

- This hemolysate contains hemoglobin for analysis.

- **7. Procedure for Cellulose Acetate Electrophoresis**
- **Step 1: Preparation of Membrane**
- Cellulose acetate strip is:
- Soaked in **electrophoresis buffer**
- Blotted to remove excess buffer

- **Step 2: Sample Application**

- Hemolysate is applied using:
- Applicator sticks
- Sample template
- Small spots are placed along the origin line.

- **Step 3: Electrophoresis**

- Strip placed in electrophoresis chamber.
- Conditions:
- Voltage: **200–300 V**
- Time: **20–30 minutes**
- Hemoglobin migrates across the membrane.

- **Step 4: Staining**
- Strip stained using:
- Common stains:
- **Ponceau S**
- **Amido black**
- Staining allows visualization of hemoglobin bands.
- **Step 5: Destaining**
- Excess stain removed using:
- Acetic acid solution
- Bands become clearly visible.

- **Step 6: Interpretation**

- Bands are compared with **control hemoglobin samples**.
- Band positions identify hemoglobin type.

- **8. Interpretation of Electrophoresis Patterns**

- **Normal Adult Pattern**

- HbA dominant
- Small HbA₂
- Trace HbF
- Pattern:
- HbA >>> HbA₂ > HbF

- **Common Abnormal Patterns**

- **1. Sickle Cell Disease**

- Pattern:

- HbS predominant
No HbA

- Example:

- $\text{HbS} > \text{HbF} > \text{HbA}_2$

- **2. Sickle Cell Trait**
- Pattern:
- HbA and HbS present
- Example:
- HbA 60%
HbS 40%

- **4. Beta Thalassemia**

- Characteristic findings:

- Increased HbA₂

- Increased HbF

- **4. Modern Alternatives**

- Advanced techniques include:

- **High Performance Liquid Chromatography (HPLC)**

- **6.3 Hemoglobin Electrophoresis**

- Used to detect **hemoglobinopathies** such as:
- Sickle cell disease
- Thalassemia
- Hemoglobin C disease
- Principle:
- Different hemoglobin variants migrate differently under an electric field.

- **6.4 Flow Cytometry for Paroxysmal Nocturnal Hemoglobinuria (PNH)**
- Used to detect **absence of CD55 and CD59 on RBCs.**
- PNH causes:
- Complement-mediated intravascular hemolysis
- Hemoglobinuria
- Thrombosis.

- **7. Bone Marrow Examination**

- Bone marrow aspiration may be performed when necessary.
- Findings in hemolytic anemia:
 - **Erythroid hyperplasia**
 - Increased erythropoiesis
 - This reflects bone marrow compensation for RBC destruction.

Chemical Tests of Haemoglobin Catabolism

- **1. Introduction**
- Hemoglobin catabolism refers to the **biochemical breakdown of hemoglobin released from destroyed red blood cells (RBCs)**. This process occurs mainly in the **reticuloendothelial system**, particularly in the **spleen, liver, and bone marrow**.
- When RBCs are destroyed during normal aging or in **hemolytic anemia**, hemoglobin undergoes several metabolic steps leading to the formation of **bilirubin, urobilinogen, and stercobilin**.
- Laboratory chemical tests that detect these products are important in:
 - Diagnosing **hemolytic anemia**
 - Monitoring **liver function**
 - Differentiating **types of jaundice**
 - Evaluating **intravascular hemolysis**
- Medical laboratory scientists perform these tests on **blood, urine, and fecal samples**.

2. Overview of Hemoglobin Catabolism

- When RBCs are destroyed:
- Hemoglobin splits into:
 - **Heme**
 - **Globin**
- Globin → broken down into **amino acids**
- Heme → converted to:
- **Biliverdin**
- **Unconjugated bilirubin**
- **Conjugated bilirubin**
- **Urobilinogen**
- **Stercobilin**
- These products form the basis of several **chemical diagnostic tests** used in hematology laboratories.

Chemical Tests for Hemoglobin Catabolism

- Important laboratory tests include:
- Serum bilirubin tests
- Urine urobilinogen test
- Urine hemoglobin test
- Hemosiderin detection
- Serum haptoglobin measurement
- Lactate dehydrogenase (LDH) assay
- Fecal stercobilinogen test
- Each test reflects a different stage of hemoglobin degradation.

- **4. Serum Bilirubin Test**
- **Principle**
- Bilirubin reacts with **diazotized sulfanilic acid** to form a **colored azobilirubin compound**, which can be measured spectrophotometrically.
- **Diazo reaction (Van den Bergh reaction)**
- Types of bilirubin measured:
- **Indirect (Unconjugated) bilirubin**
- **Direct (Conjugated) bilirubin**
- **Total bilirubin**

- **Laboratory Method**
- Common methods include:
- **Diazo reaction (Van den Bergh reaction)**
- Spectrophotometric analysis

Condition

Hemolytic anemia

Obstructive jaundice

Liver disease

Bilirubin Finding

Increased **unconjugated bilirubin**

Increased **conjugated bilirubin**

Mixed increase

- In **hemolytic anemia**, excessive RBC destruction produces large amounts of **unconjugated bilirubin**, leading to **jaundice**.

- **6. Urine Hemoglobin Test**

- **Principle**

- Free hemoglobin in urine reacts with **peroxidase reagents**, producing a color change.
- Common screening tests include:
 - **Benzidine test**
 - **Orthotolidine test**
 - **Dipstick reagent strip**

- **Significance**
- Hemoglobinuria indicates:
- **Severe intravascular hemolysis**
- Acute transfusion reactions
- Paroxysmal nocturnal hemoglobinuria
- Malaria
- Urine may appear **reddish or dark brown.**

- **7. Hemosiderin Detection Test**

- **Principle**

- Hemosiderin is detected using **Perl's Prussian blue reaction**, where ferric iron reacts with potassium ferrocyanide to form a **blue pigment**.

- **Method**

- Centrifuge urine sample.
- Prepare sediment smear.
- Stain using **Prussian blue stain**.

- Presence of **blue granules in renal tubular cells** indicates **hemosiderinuria**.
- **Clinical Significance**
- Hemosiderinuria indicates:
- **Chronic intravascular hemolysis**
- Seen in:
 - Paroxysmal nocturnal hemoglobinuria
 - Severe hemolytic anemia
 - Mechanical hemolysis

- **8. Serum Haptoglobin Test**

- **Principle**

- Haptoglobin is a **plasma protein that binds free hemoglobin** released during hemolysis.
- The **hemoglobin-haptoglobin complex** is rapidly cleared by the liver.

- **Laboratory Measurement**
- Measured using:
- **Immunoturbidimetric methods**
- **Nephelometry**

- **9. Lactate Dehydrogenase (LDH) Assay**

- **Principle**

- LDH is an intracellular enzyme released when RBCs are destroyed.
- It catalyzes conversion of **lactate to pyruvate**.

- **10. Fecal Stercobilinogen Test**

- **Principle**

- Stercobilinogen is formed when intestinal bacteria metabolize bilirubin.
- It is oxidized to **stercobilin**, which gives stool its **brown color**.